Ultrafast carrier capture in InGaAs quantum posts

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To explore the capture dynamics of photoexcited carriers in semiconductor quantum posts, optical pump terahertz (THz) probe and time-resolved photoluminescence spectroscopy were performed. The results of the THz experiment show that after ultrafast excitation, electrons relax within a few picoseconds into the quantum posts, which act as efficient traps. The saturation of the quantum post states, probed by photoluminescence, was reached at approximately ten times the quantum post density in the samples. The results imply that quantum posts are highly attractive nanostructures for future device applications. © *2009 American Institute of Physics*. [doi:10.1063/1.3275666]

Semiconductor nanostructures such as quantum dots (QDs) have been incorporated in the design of optoelectronic devices[1] and have provided a rich platform for investigating three-dimensionally confined carriers in a tailorable environment.[2,3] Self-assembled nanostructures such as InGaAs QDs and quantum wires are attractive for low threshold lasers and sensitive detectors due to their strong confinement, and their ultrafast carrier dynamics have been studied intensely over the last decade.[4–7] In particular for laser devices, the carrier capture dynamics into the active region play a crucial role for performance and efficiency. Recently, quantum posts (QPs)—nanowirelike structures that are terminated by QDs on each end—have been introduced[8] and integrated into interband lasers, showing remarkably low threshold currents.[9] While for interband devices the capture dynamics of the electrons and holes are equally important and can be studied by time-resolved photoluminescence (TRPL), it is of large interest to study the electron capture separately for possible unipolar device applications.

In this letter, we report on the carrier capture dynamics in InGaAs QPs after photoexcitation into a surrounding GaAs matrix. We study these dynamics by optical pump terahertz (THz) probe (OPTP) experiments and TRPL. In the case of OPTP, the probe frequencies were well below any resonance in the system and therefore probed only the ac conductivity of the samples, which is dominated by electrons in the conduction band.[11–13] After excitation into the three-dimensional and two-dimensional (2D) matrices that surround the QPs, the temporal decay of the ac conductivity directly corresponds to capture into the QPs, where due to nanometer confinement the ac conductivity drops[12] significantly and only small polarizabilities give a response to the THz field.[14] TRPL measurements served as a probe to determine at which excitation density the QP states are saturated—allowing us to deconvolve the different timescales involved in the OPTP results.

QPs represent self-assembled, height controlled nanorods grown by molecular beam epitaxy that are embedded vertically in GaAs and horizontally in an InGaAs matrix.[8] This is schematically illustrated in Fig. 1. The entire structure is coherently strained and PL lines characteristic of the QPs, the energetically higher InGaAs matrix, as well as the GaAs host material can be observed separately, depending on the excitation density. Since even at high QP densities the InGaAs matrix still resembles a structurally intact quantum well (QW), whose width matches the height of the QPs, it is essential to compare data of QP samples with those of a reference QW structure that does not contain QPs. In the work presented here, we have studied two QP samples with QP heights of 40 and 50 nm and an areal density of 1×10^{10} cm^{-2}, as well as a 40 nm QW sample where the QW composition matches the InGaAs matrix that surrounds the QPs. All samples consisted of two periods with a 400 nm

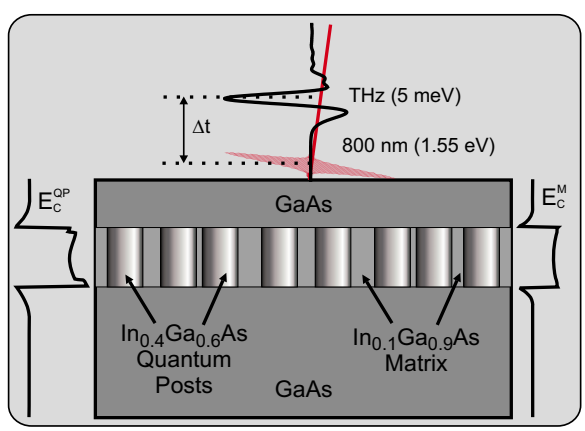

FIG. 1. (Color online) Schematic of the samples and experimental setup. The QPs are vertically embedded in a GaAs matrix, while they are laterally surrounded by a QW-like $In_{0.1}Ga_{0.9}As$ matrix. The conduction band profiles along the posts E_c^{QP} and the matrix E_c^{M} are shown schematically to the left and the right of the sample, respectively. Since the QPs are terminated by a QD at each end, their energy profile is modulated by strain. Also shown is the experimental configuration, where the near-infrared pump pulse and the THz probe pulse are normally incident onto the sample. Δt is the time delay between the pump and probe pulses.

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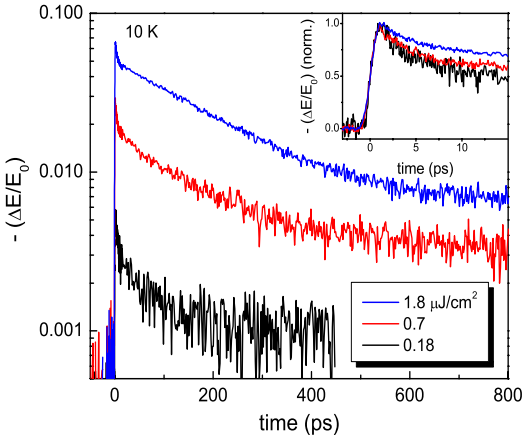

FIG. 2. (Color online) Results for the optical pump THz probe experiment on the QP sample for three different pump fluences at 10 K. The inset shows the normalized change in electric field on a short timescale.

thick GaAs buffer in between. PL spectra taken on the QP samples at a temperature of 10 K show broad (full width at half maximum 50 nm) emission around 1 μm whereas the QW shows PL at 890 nm with a width of 5 nm.

In order to study only the dynamic behavior due to the *capture* by the QP layers, the optical pump beam and the THz beam are normally incident on the sample. In this geometry, the allowed transitions are defined by the *radial* confinement. At a diameter of typically 20 nm, these transitions are calculated to be around 25 meV, sufficiently above our probe frequencies.[10] This ensures that no resonances are probed. The *vertical* confinement actually provides resonances in the 5–20 meV range, depending on the height of the QPs. Probing these transitions, however, would require an electric field at the QPs that has a strong component in the growth direction, similar to the intersubband absorption in wide QWs (Ref. 15) and is not studied in this work.

In Fig. 1, the OPTP experiment is schematically illustrated. The QP and QW samples are excited with 50 fs near-infrared light pulses centered at 800 nm that are delivered from a regenerative Ti:Sa amplifier running at 250 kHz, whose output is split into following three pulses: one for excitation of the sample, one for THz generation and one for electro-optic sampling of the THz pulse. The THz pulses are generated by optical rectification in a 500 μm thick ⟨110⟩-oriented ZnTe crystal. The THz radiation is collimated and refocused by off-axis parabolic mirrors onto the sample and the electro-optic crystal, and another 500 μm thick ⟨110⟩ ZnTe crystal is used for detection. This system provides a large dynamic range for the sampled THz electric field due to the high repetition rate while simultaneously providing sufficient pulse energy to optically excite the samples in a wide range of excitation densities. Under identical fluences, TRPL was recorded from the samples employing a Ti:Sapphire oscillator in combination with a streak-camera.

To explore the capture dynamics, we performed one-dimensional pump-probe scans where only the peak electric field of the THz pulse was monitored while the pump-probe delay (see Fig. 1) was varied. Being sufficiently detuned from a resonance, the change of the electric field then reflects the pump-induced conductivity change $\Delta\sigma(t)$.[12] This was confirmed by taking THz field scans at intervals of 100 ps of pump probe delay, where no change in the overall shape or

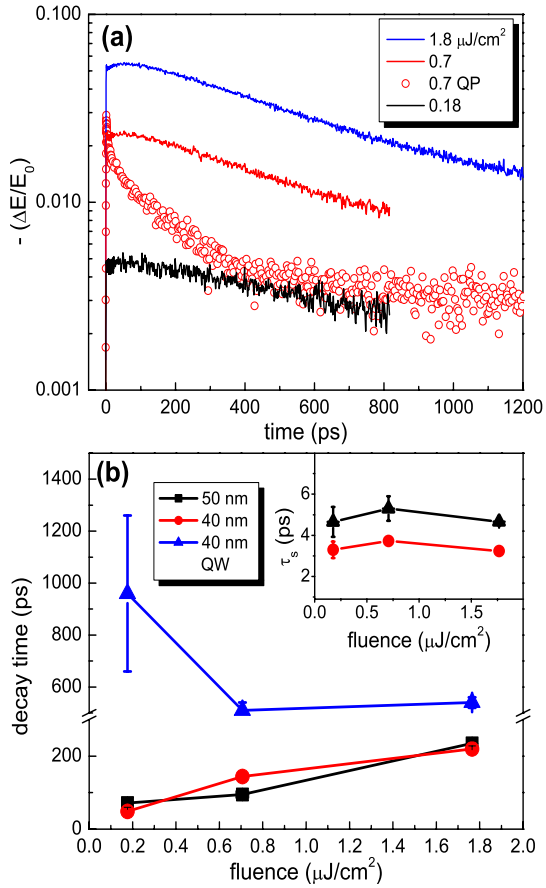

FIG. 3. (Color online) (a) Results for the optical pump THz probe experiment on the QW sample for three different pump fluences at 10 K. The lower graph (b) shows the extracted decay times for the QW (blue) and the QP samples (black and red) at different pump fluences. The inset in (b) shows the fast decay component, followed by the intermediate decay (see text).

phase of the induced field changes was observed. In contrast to TRPL, here only the electron capturing process is observed since the hole contribution to the conductivity is negligible. In Fig. 2 we present the OPTP results of the 50 nm QP sample, taken for three different pump fluences. At the lowest pump fluence of 0.18 μJ cm^{-2}, the excitation density is approximately 2×10^{10} cm^{-2} which is only a factor of 2 above the QP density of the samples. Hence, this system allows measurement of the capture process without saturation of the QP states. For all fluences, we observe a sharp rise of the transient which reflects the generation of free carriers in the GaAs matrix.[16] The observed rise time is limited by the duration of the probe pulse and cannot be determined. Immediately after the peak, the transient drops quickly within a few picoseconds (see inset of Fig. 2) and then significantly slows down (tens of picoseconds) and eventually slows down even more. As the optical pump fluence is increased, the intermediate decay becomes longer.

To determine which of these observed processes are caused by the presence of QPs in the sample, we performed identical measurements in a QW structure that has the same composition as the matrix surrounding the QPs, i.e., an $In_{0.1}Ga_{0.9}As$ QW with a thickness of 40 nm. Carriers were generated in the GaAs host material. The results for the QW are shown in Fig. 3(a). After photoexcitation, the signal immediately rises as in the QP sample. However, no short decay times are observed here and in fact the signal is first

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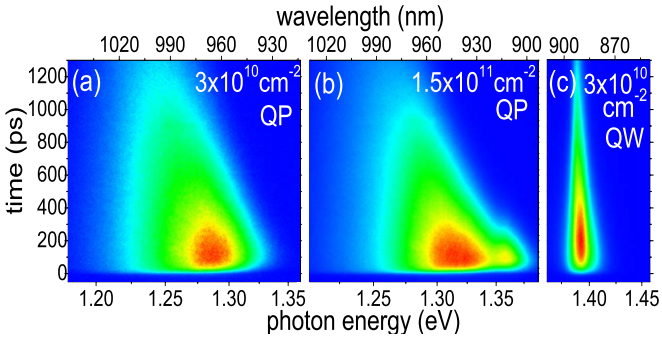

FIG. 4. (Color online) TRPL (false color plot) taken at 10 K from the 40 nm QP sample at excitation densities of 3×10^{10} cm^{-2} (a) and 1.5×10^{11} cm^{-2} (b) and the 40 nm QW sample (c). The time resolution of the streak camera was 15 ps.

constant and then even *rises* up to a delay time of about 100 ps. After the maximum, the signal decays with a time-constant of several hundreds of picoseconds due to interband recombination. In contrast to the QP sample, the dynamics change only slightly with increasing pump fluence. This behavior is in strong contrast to what we observed in the QP sample, which is plotted again for the intermediate fluence in Fig. 3(a) as the red open dots; while the initial change of the electric field is about the same and corresponds to the excitation of free carriers in the GaAs material, the dynamics that follow are strikingly different.

The time constants extracted from the measurements with biexponential fits are shown in Fig. 3(b). The main graph displays the long time constants observed in the QW and the QP samples. While for the QW sample no clear dependence on the fluence can be extracted, we see an increase in the intermediate decay time for the QP samples. This component corresponds to free carrier absorption in the $In_{0.1}Ga_{0.9}As$ matrix and can be interpreted as the onset of a saturated QP population—and was observed in QDs. [13]

Figure 4 shows TRPL measurements, which further support this picture; at low excitation density, only QP states contribute to the luminescence which shifts toward lower energy with increasing time delay due to the relaxation of electrons and holes within the QPs. Similar PL dynamics and energy shifts have been observed in QDs.[17] At a density that matches the intermediate density of the OPTP experiments, a second, narrower line appears at around 1.36 eV (913 nm). That line represents the matrix, indicating that the QP states are filled. It decays within a few hundred picoseconds, much faster than the PL taken from the QW sample shown in Fig. 4(c). The fact that the energies of the QW PL and the QP matrix PL do not match indicates a different In concentration for these two regions. Increasing the fluence on the QP sample (not shown), however, showed that this line does not saturate and finally dominates the entire PL spectrum which is the expected behavior of a QW-like system.

The initial decay—shown in the inset of Fig. 3(b)—however, does not depend on the excitation density. While its contribution to the overall decay decreases with rising fluence, it merely remains constant. This fact and the absence of this fast decay channel in the QW sample indicates that this time reflects the capturing of electrons from the GaAs materials directly into the QPs. Interestingly, this capture time differs for the two different QP sizes investigated here. However, the origin of this difference is not clear, and more sizes have to be investigated to draw a clear conclusion, which is beyond the scope of this paper. The capture time of 3–5 ps is similar to what has been observed for QDs, where in contrast to the conductivity, mid-infrared transitions in the conduction band of the sample were probed[18] in a comparable excitation regime. There, it was shown that relaxation into the QDs involves the 2D wetting layer states. The question to what degree the matrix is involved in the QP capture process cannot be fully answered from our experiments. However, the facts that in the TRPL the QP and matrix PL appear simultaneously and that the latter [see Fig. 4(b)] decays much faster than in the QW sample indicate that relaxation is not only possible directly into the QP but also via the 2D matrix.

In conclusion, we have investigated the carrier capture dynamics in QP heterostructures. Direct comparison to QW samples has shown that QPs act as efficient electron traps after optical excitation, and a capture time of a few picoseconds was determined. The results demonstrate the feasibility of using QPs as active elements in future devices where carriers in the bulk must be efficiently transferred into the QPs.

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